Spatial and Spectral Structure Preserved Self-Representation for Unsupervised Hyperspectral Band Selection

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***Abstract*— As an effective manner to reduce data redundancy and processing inconvenience, hyperspectral band selection aims to select a subset of informative and discriminative bands from the original data cube. Although a large number of approaches have been proposed and obtained great success, they still face at least two issues. First, most of the previous methods only consider the redundancy between neighbor bands, while the global information has been ignored. Second, each band is often treated as a whole and reshaped to a feature vector without considering the spatial structure of different regions. In this article, in order to address these issues, we propose a spatial and spectral structure preserved self-representation model for unsupervised hyperspectral band selection without using any label information, referred to as S^{4}P briefly. Different from previous methods that stretch each band into a feature vector, the first principal component of the original hyperspectral cube is segmented into different superpixels, which can reflect the spatial structure of homogeneous regions. Then each band can be represented by a superpixel-level feature vector, and the self-representation model is utilized to learn the spectral correlation of different bands. In addition, an adaptive and weighted multiple graph fusion term is designed to generate a unified similarity graph between different superpixels, which is used to capture the spatial structure in the self-representation space. Finally, an $l_{2,1}$-norm is imposed on the self-representation coefficient matrix to measure the band importance. We design an alternative update scheme to optimize the resultant problem, the self-representation coefficient matrix and the superpixel-wise similarity graph can boost each other during the updating process to obtain optimal results. Extensive experiments with detailed analysis of three public datasets are conducted to validate the superiority of the proposed S^{4}P when compared with other state-of-the-art competitors.**

***Index Terms*— Clustering, graph fusion, hyperspectral band selection, similarity graph learning.**

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I. Introduction

OWING to the rapid progress of hyperspectral imaging technology, a tremendous number of hyperspectral images (HSIs) can be captured by hyperspectral sensors which record the spectral responses of objects to a set of narrow electromagnetic bands with different wavelengths [1], [2]. Due to richer discriminative physical clues from different spectral bands, HSIs attract growing attention in recent decades and are widely used in various fields, such as vegetation coverage estimation [3], environmental monitoring [4], mineral exploration [5], and landslide detection [6]. However, the high dimensionality causes at least two issues for HSI data processing. First, a significant computational burden would hinder the efficiency of many learning models. Second, there are unavoidable redundant/noisy bands mixed in the original hyperspectral data cube, which inevitably degenerates the final processing performance. Therefore, dimension reduction for HSI data is an urgent task.

In the past decades, many approaches have been put forward to reduce the dimensionality of HSI data cube, these methods can be mainly categorized into two classes, i.e., feature extraction and band selection [7]. Feature extraction methods aim to project original high-dimensional data into a new feature space with a lower dimensionality. By regarding each band as a feature dimension, traditional feature extraction methods such as principal component analysis (PCA) [8], [9], independent component analysis (ICA) [10], [11], linear discriminant analysis (LDA) [12], and maximum noise fraction [13] can be used for HSI data. However, the physical meaning of each band is lost in the new feature space. As to band selection, the core idea is to select a band subset from the original bands. The selected bands in the band subset are often required to be more representative and discriminative for the following specific tasks, as well as the inherent band properties and physical meaning being well preserved. As a result, band selection attracts more attention. In order to retain the feature interpretability, both band selection and spatial noise reduction are also integrated to extract low-dimensional spectral–spatial feature representation of HSI [14]. Since the

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representative bands could not be incorrectly selected due to water absorption and/or low signal-to-noise, after grouping neighboring bands, Fu et al. [14] developed a modified band normalization matched filter ranking strategy to select more representative bands, thereby the effect of noise can be effectively mitigated. In this article, our proposed method also falls into band selection [15], [16].

Based on the availability of data labels, previous band selection methods can be roughly classified into supervised ones [17], [18], [19] and unsupervised ones [20], [21], [22] based on the availability of data labels. As to supervised methods, it is critical to train a learning model while designing an appropriate model is difficult. In addition, obtaining sufficient data labels is not easy and also laborious. Compared to supervised methods, unsupervised band selection is more flexible since it works without using data labels which are unavailable in most practical cases. Without data labels, some prior information like the local geometric structure of original HSI data is often used in unsupervised band selection methods with certain criteria such as band similarity [23], information divergence [24], maximum ellipsoid volume [25], and structural similarity [26]. Three main strategies include ranking, searching, and clustering are often used for unsupervised hyperspectral band selection. As to ranking-based methods, they often consist of two steps, i.e., band importance evaluation and discriminative band selection. Some typical criteria such as maximum variance [8], information entropy [27], information divergence [28], and structural similarity [26] are usually used for evaluating the importance of different bands. As to searching-based methods, the desired bands are selected by certain greedy or evolutionary algorithm such as immune clone, determinantal point processes [29], discrete gravitational search algorithm [15], and particle swarm [30]. For clustering-based methods, original bands are first partitioned into different groups and the most representative band of each group is then picked out to constitute the final band subset, such as [16], [24], [31], [32], [33], [34], [35], [36], [37], [38], [39], and [40]. Furthermore, to reduce the influence of synonyms spectrum or foreign objects with the same spectrum, many superpixel-based band selection methods are proposed in recent years. Zhang et al. [41] exploited a superpixel-based similarity graph to capture the spectral and spatial information from the segmented HSIs. This kind of method often obtains satisfying results since the specific and common information of HSIs can be well captured according to the similarity graph learning [42], [43], [44], [45], [46], [47], [48], [49]. In addition, by considering the noise influence on the band subset determination, Sun et al. [50] proposed an adaptive distance-based band hierarchy clustering model to acquire flexible number of band subset by leveraging a tree hierarchy-based data structure of original HSIs, with the effect of noisy bands can be effectively avoided. Similarly, a hyperbolic clustering-based band hierarchy model [51] is explored to discover the underlying spectral structure of HSIs and more consistent band selection results can be obtained. Since the desired band subset is obtained from the band clusters, the performance of clustering methods would directly affect the final band selection results. Although there are prosperous clustering methods proposed for unsupervised band selection, they still face at least two issues. First, most of the previous methods only consider the redundancy between neighbor bands, while the global information has been ignored. Second, each band is often treated as a whole and reshaped to a feature vector without considering the spatial structure of different regions.

It should be noted that there are also many deep neural network-based band selection methods being proposed in the past few years. Among these networks, the deep auto-encoder is the most commonly used backbone for feature representation learning [52], [53], [54], [55], [56], [57], [58]. Given original HSIs, these methods often use the deep auto-encoder to learn the band weight coefficients for selecting those important bands, then original bands can be reconstructed by the selected band subset via the corresponding decoder with a reconstruction loss function. Due to the powerful feature learning capability, deep neural network-based methods often get remarkable performance. However, one needs to carefully design the network structure and training loss functions, and the interpretability of the learning process is not clear. Therefore, we still focus on clustering-based unsupervised hyperspectral band selection.

In this work, in order to address the above issues that existed in previous clustering-based models, we introduce a spatial and spectral structure preserved self-representation model (S^4P) for unsupervised hyperspectral band selection, as shown in Fig. 1. The first principal component of the original HSI cube is segmented into different superpixels, which can reflect the spatial structure of homogeneous regions. Then the self-representation model is deployed to learn the spectral correlation of different bands at a superpixel level. In addition, we design an adaptive and weighted multiple graph fusion term for learning a unified similarity graph between different superpixels for spatial structure preservation in the self-representation space. Finally, an $l_{2,1}$-norm is imposed on the representation coefficient matrix to measure the band importance. In our proposed S^4P, spectral correlation among different bands as well as the similarity between different regions within each band are taken into consideration for learning band importance. The main contributions of this work can be summarized as follows.

1) We propose a self-representation model-based unsupervised hyperspectral band selection method that fully takes both the spatial and spectral structure of the original hyperspectral cube into consideration.
2) We introduce an adaptive and weighted multiple graph fusion term to learn a unified similarity graph between different regions of the 2-D hyperspectral slice for capturing spatial structure information in the spectral self-representation space.
3) We design an alternative update scheme to optimize the resultant problem, and extensive experimental results with detailed analysis of different public datasets demonstrate the superiority of the proposed S^4P when compared with other competitors.

II. Prior Works

In this section, we will give a brief introduction to the self-representation model for hyperspectral band selection. Before that, the commonly used notations are presented in

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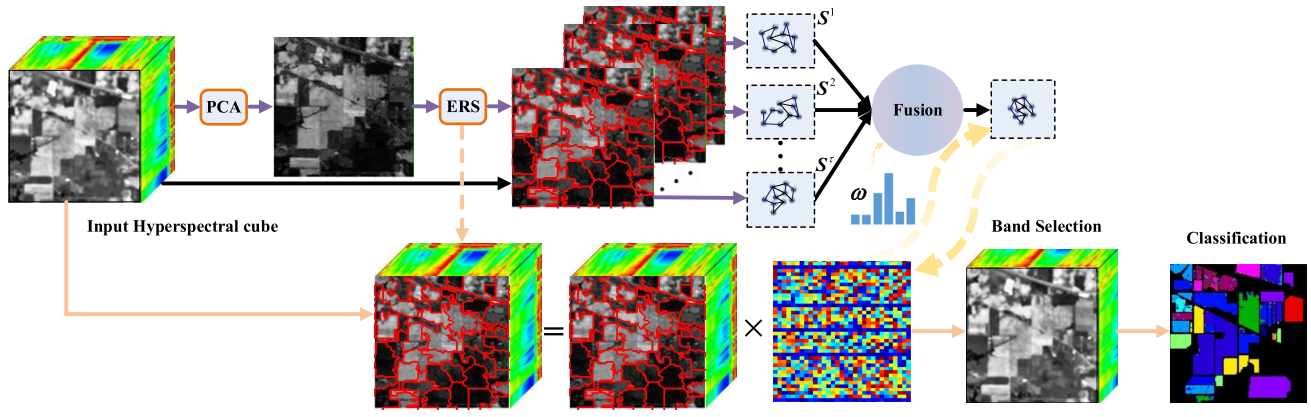


Fig. 1. Flowchart of our hyperspectral band selection method via spatial and spectral structure preserved self-representation model. First, in order to capture the spatial structure of homogeneous regions of the hyperspectral scene, the first principal component of the original hyperspectral cube is segmented into different superpixels. Then, each band is represented by a superpixel-level feature vector and the self-representation model is utilized to learn the spectral correlation of different bands. In order to preserve the spatial structure in the self-representation space, an adaptive and weighted multiple-graph fusion term is designed to generate a unified similarity graph between different superpixels. As to the self-representation coefficient matrix, the $l_{2,1}$-norm is imposed for measuring the importance of different bands.

the following. Given a HSI cube $\mathbf{Q} \in \mathbb{R}^{H\times W\times B}$, where H and W represent the spatial width and height of a certain scene, respectively. B denotes the number of bands. Considering that each band can be represented by a combination of other bands, the self-representation model aims to obtain the optimal coefficient matrix to encode the correlation of all hyperspectral bands. For ease of processing 3-D HSIs, a commonly used method is stretching the original $\mathbf{Q}$ into a set of 2-D slices $\mathbf{F} \in \mathbb{R}^{HW\times B}$ along with the spectral response dimension, and then the band selection problem can be formulated as

$$\min_{\mathbf{Z}} \Phi(\mathbf{F}, \mathbf{FZ}) + \beta\Psi(\mathbf{Z}) \tag{1}$$

where $\Phi(\cdot)$ and $\Psi(\cdot)$ denote the loss function and regularization term, respectively, and β is a parameter. For the loss function term, $\Phi(\cdot)$ is often constructed in the form of the Frobenius norm to measure the reconstruction error. As to the $\Psi(\cdot)$, many existing self-representation models often adopt diverse regularization terms to obtain the different properties of the coefficient matrix $\mathbf{Z}$, such as $||\mathbf{Z}||_F^2$ [59], $||\mathbf{Z}||_{2,1}$ [60], and $||\mathbf{Z}||_*$ [61].

After generating the coefficient matrix $\mathbf{Z}$, the k-means clustering is conducted on the corresponding similarity matrix $(|\mathbf{Z}| + |\mathbf{Z}|^\top/2)$ to divide all bands into multiple clusters, from which the optimal band subset is selected.

III. Proposed Method

In many previous clustering-based band selection methods, different pixel values that correspond to reflection characteristics of different objects in each band are often stacked to generate a single feature vector for post-processing. However, this operation would raise two problems. First, the stacked feature vector will be with high dimensionality. Taking a certain band with size 145×145 in the Indian Pines dataset, for example, the dimension of each feature vector is 21 025. However, the number of the bands in this dataset is only 224, which will cause the "curse of dimensionality" problem. In addition, the high dimensionality also increases the computational burden for some learning algorithms. Second, the spatial structure of different regions in each band is also ignored. Therefore, we segment each band into different homogeneous regions, and the spatial distribution of different regions is taken into consideration for learning the correlation between different bands. Owing to the good property for capturing both spatial and spectral information of HSIs, superpixel segmentation is widely used in hyperspectral processing, such as classification [62], [63], [64], unmixing [65] and restoration [66].

A. Homogeneous Region Generation

As to a certain scene, there are usually various objects with different reflection characteristics. Therefore, the same object should share similar spectral values, while different objects should be with different spectral values. In order to effectively characterize and utilize this kind of spatial semantic information. We segment each band into several homogeneous regions with each region representing a certain part of an object. For achieving a balance between efficacy and efficiency, the entropy rate superpixel segmentation (ERS) method [67] is used to generate multiple homogeneous regions. Since the scene represented from different spectral bands are the same, in order to reflect the spatial structure of the scene more clearly, we implement PCA [68] on the original hyperspectral cube and obtain the first principal component (denoted as P) of the hyperspectral. Then, we can generate N superpixels by performing ERS on P as follows:

$$P = \bigcup_{i}^{N} \mathcal{R}_i, \quad \text{s.t.} \quad \mathcal{R}_i \cap \mathcal{R}_j = \emptyset, \quad (i \neq j) \tag{2}$$

where $\mathcal{R}_i$ is the pixel set of the ith superpixel. Then we map the superpixel segmentation to different bands with the same segmentation label.

B. Formulation of S^4P

Our proposed S^4P is based on the typical self-representation model that assumes that each band can be linearly reconstructed from other bands. For simplicity, we use the basic

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linear form. Of course, we believe that other nonlinear formulations could obtain better performance. After we segment each band into N superpixels, each superpixel can be represented by the averaged spectral values of pixels located in it. As a result, each band can be represented by an $N \times 1$ feature vector. In such a manner, the computational cost of the following optimization algorithm will be significantly reduced since $N \ll W \times H$. In addition, the spatial structure of different superpixels will be also easily embedded into our proposed model. Specifically, let $\mathbf{X} \in \mathbb{R}^{N \times B}$ denote the hyperspectral data represented by superpixel level feature vectors. Then the representation model of the ith band can be written as follows:

$$\mathbf{x}_i = \mathbf{X}\mathbf{z}_i \quad \text{s.t.} \quad \varphi(\mathbf{z}_i) \tag{3}$$

where $\mathbf{x}_i$ denotes the ith column of $\mathbf{X}$, and $\mathbf{z}_i$ denotes the ith column of the representation coefficient matrix $\mathbf{Z}$. $\varphi(\mathbf{z}_i)$ is a certain regularization imposed on $\mathbf{z}_i$.

Considering all the bands together, we get the following matrix form:

$$\mathbf{X} = \mathbf{X}\mathbf{Z}, \quad \text{s.t.} \quad \varphi(\mathbf{Z}) \tag{4}$$

where $\mathbf{Z} \in \mathbb{R}^{B \times B}$ is the coefficient matrix with its ith row expressing the representation capability of original ith band for reconstructing other bands. Therefore, commonly used $l_{2,1}$-norm can be imposed on $\mathbf{Z}$ to constrain its row sparsity.

However, (4) only uses the spectral correlation between different bands, while the spatial structure of a certain scene is ignored. It is a rational assumption that if two regions are similar in the original data space, they should be also similar in the reconstructed space, which can be written as the following optimization problem:

$$\min_{\mathbf{Z}} \frac{1}{2} \sum_{i,j=1}^{N} ||\mathbf{x}^i\mathbf{Z} - \mathbf{x}^j\mathbf{Z}||_2^2 S_{ij} \tag{5}$$

where $\mathbf{x}^i$ and $\mathbf{x}^i$ represent the ith and jth row of $\mathbf{X}$, and S_{ij} is the similarity between the ith and jth superpixel region. By a simple algebra, (5) can be written as the following matrix form:

$$\min_{\mathbf{Z}} Tr(\mathbf{Z}^\top \mathbf{X}^\top \mathbf{L}\mathbf{X}\mathbf{Z}) \tag{6}$$

where $\mathbf{L} \in \mathbb{R}^{N \times N}$ is the corresponding Laplacian matrix with $\mathbf{L} = \mathbf{D} - \mathbf{S}$, and $\mathbf{D}$ is the diagonal degree matrix with $\mathbf{D}_{ii} = \sum_j S_{ij}$.

By now, the question is how to calculate $\mathbf{S}$? A direct and simple way is to calculate the similarities between different superpixels of the first principle component of original HSIs by using commonly used distance metrics such as Gaussian kernel function and cosine similarity. However, the information encoded in the first principle component of original HSIs is not complete. Therefore, we use the first τ principle components to get a more robust similarity graph via a graph fusion strategy. In our experiments, when $\tau = 3$, we can already get good results. With the increased number of principle components, there is no obvious performance gain. Therefore, on the premise of ensuring good performance, we do not use more principal components for reducing computational costs. For each principle component from the τ ones, it describes the scene from a certain specific aspect, we construct a similarity graph of different superpixels, and then multiple graphs are aggregated to generate a unified graph $\mathbf{S}$ which is more accurate to reflect the correlation of different superpixels. In detail, we introduce the following unified graph learning term:

$$\begin{aligned} &\min_{\mathbf{S}} \sum_{v=1}^{\tau} \boldsymbol{\omega}_v ||\mathbf{S} - \mathbf{S}^v||_F^2, \\ &\text{s.t.} \quad S_{ij} \geq 0, \mathbf{s}_i^\top \mathbf{1}_N = 1, \boldsymbol{\omega}^\top \mathbf{1}_\tau = 1 \end{aligned} \tag{7}$$

where $\mathbf{1}_N$ is a N-dimensional all-one-column vector, and $\mathbf{1}_\tau$ has the same definition. $\mathbf{S}^v$ represents the similarity graph constructed from the vth principle component of original HSIs. Since different principle components could contribute differently to learning the unified similarity graph, we add a weight $\boldsymbol{\omega}_v$ for every single graph and it can be adaptively learned during the optimization process. Different from many previous methods which construct the similarity graph by treating each band as a single feature vector and only using the spectral information, here we aim to construct the similarity graph for different spatial regions. It is an intuitive fact that if two superpixels are with similar spectral values, they are more likely to represent the same objects and therefore should be similar to each other.

For simplicity, $\mathbf{S}^v$ is calculated by using the Gaussian kernel function defined as follows:

$$S_{ij}^v = \exp\left(\frac{||p_i^v - p_j^v||^2}{-2\sigma^2}\right) \tag{8}$$

where p_i^v and p_j^v represent the averaged spectral value of ith and jth superpixel in the vth principle component.

By combining (4), (6), and (7) together, and imposing the row sparsity constraint on $\mathbf{Z}$, we get our S^{4}P model as follows:

$$\begin{aligned} \min_{\mathbf{Z},\boldsymbol{\omega}_v,\mathbf{S}} &||\mathbf{X} - \mathbf{X}\mathbf{Z}||_F^2 + \sum_{v=1}^{\tau} \boldsymbol{\omega}_v ||\mathbf{S} - \mathbf{S}^v||_F^2 \\ &+ \alpha Tr(\mathbf{Z}^\top \mathbf{X}^\top \mathbf{L}\mathbf{X}\mathbf{Z}) + \gamma ||\mathbf{Z}||_{2,1} \\ \text{s.t.} \quad &S_{ij} \geq 0, \mathbf{s}_i^\top \mathbf{1}_N = 1, \boldsymbol{\omega}^\top \mathbf{1}_\tau = 1 \end{aligned} \tag{9}$$

where the $l_{2,1}$-norm imposed on matrix $\mathbf{Z}$ is defined as $||\mathbf{Z}||_{2,1} = \sum_{i=1}^{B} ||\mathbf{z}^i|| = \sum_{i=1}^{B} (\sum_{j=1}^{B} {Z_{ij}}^2)^{1/2}$. As can be seen from (9), our S^{4}P model integrates band correlation learning, spatial region similarity graph fusion, and principle component weight learning into a unified framework. During the following optimization process, different variables will be promoted by each other to reach optimal results. Although several advanced hyperspectral band selection methods have also considered the spectral and spatial structure of original data, e.g., Jia et al. [69] proposed a multiscale superpixel-level group clustering framework (MSGCF), which searches the maximum points of superpixel-level KL-l_1 distance of adjacent bands for separating bands into groups. In order to learn high-level deep semantic features, Xu et al. [70] designed an end-to-end spectral–spatial semantic feature learning network for unsupervised deep semantic feature extraction from HSIs. It should be noted that there are several merits of our proposed model when compared to other previous methods as follows.

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1) Since we segment each band into N superpixels, and then each superpixel can be represented by the averaged spectral values of pixels located in it so that each band can be represented by an $N \times 1$ feature vector. To this end, the computational cost of the following optimization algorithm will be significantly reduced since $N \ll W \times H$.
2) The first term in (9) is actually a spectral recombination form, i.e., the reconstructed feature vectors are just a linear combination of original spectral features. Therefore, the spectral structure of original bands can be effectively exploited.
3) For HSIs, it is a rational assumption that if two regions are similar in the original data space, they should be also similar in the reconstructed space, which is formulated by (6). The first term in (9) is actually a spectral recombination form, i.e., the reconstructed feature vectors are just a linear combination of original spectral features. Therefore, the spectral structure of original bands can be effectively exploited. Different from previous methods that treat each band as a whole by using (4), we regard each superpixel along the band dimension as shown by (5). In such a manner, the spatial structure information can be effectively exploited to learn the band correlation in the final model. In addition, different from previous methods that directly group original bands into different clusters in superpixel-level, we aim to learn the correlation of different bands and perform band clustering by using the correlation matrix. During the learning process, the local geometrical structure of different superpixels is integrated into the model by using (5). In addition, considering that only one band cannot accurately reflect the similarity between different spatially distributed superpixels due to the inconsistent quality of hyperspectral bands, we design a principle component weight learning strategy to fuse the superpixel similarity graphs constructed from several principle bands, which can boost the quality of superpixel similarity graph for local structure preservation in the self-representation model. Therefore, the proposed adaptive and weighted multiple graph fusion term aims to learn a high-quality similarity graph between different superpixels of the 2-D hyperspectral slice, which effectively captures the spatial structure information in the spectral self-representation space.
4) Rather than directly preserving the band correlation, the third term of (9) aims to preserve the spatial correlation of different objects in the reconstructed spectral self-representation space, which can capture the spatial structure to boost the spectral feature correlation matrix learning.
5) By embedding the principle component weight learning into the model, more complete information on original HSIs can be exploited to learn a more accurate unified spatial region similarity graph.

By solving (9), the importance of each band can be measured via the l_2-norm of each row of $\mathbf{Z}$.

C. Model Optimization

In order to optimize the three variables (i.e., $\mathbf{Z}$, $\boldsymbol{\omega}_v$, and $\mathbf{S}$) involved in (9), we design an iterative algorithm to solve the resultant problem. For each time, we update one of the variables with the other ones fixed. The detailed updated rules are as follows.

1) Fix $\mathbf{S}$ and $\boldsymbol{\omega}$, Update $\mathbf{Z}$: When $\mathbf{S}$ and $\boldsymbol{\omega}$ are fixed, updating $\mathbf{Z}$ is equal to solve the following problem:

$$\min_{\mathbf{Z}} ||\mathbf{X} - \mathbf{X}\mathbf{Z}||_F^2 + \alpha Tr(\mathbf{Z}^\top \mathbf{X}^\top \mathbf{L}\mathbf{X}\mathbf{Z}) + \gamma ||\mathbf{Z}||_{2,1}. \tag{10}$$

It should be noted that (10) is convex, but the $l_{2,1}$-norm regularization is nonsmooth. In this section, we design an iterative reweighted least-squares (IRLS) algorithm to solve it. Supposing the current solution of $\mathbf{Z}$ in (10) is $\mathbf{Z}_t$, we define a diagonal weighting matrix $\mathbf{G}_t$ by $g_t^i = 1/2\left\|\mathbf{z}_t^i\right\|_2$, and then $\mathbf{Z}_{t+1}$ is updated by solving the following weighted least squares problem:

$$\mathbf{Z}_{t+1} = \arg\min_{\mathbf{Z}} \begin{Bmatrix} \mathrm{Tr}((\mathbf{X}-\mathbf{X}\mathbf{Z})^\top(\mathbf{X}-\mathbf{X}\mathbf{Z})) \\ +\gamma \mathrm{Tr}(\mathbf{Z}^\top \mathbf{G}_t \mathbf{Z}) + \alpha \mathrm{Tr}(\mathbf{Z}^\top \mathbf{X}^\top \mathbf{L}\mathbf{X}\mathbf{Z}) \end{Bmatrix}. \tag{11}$$

Taking the derivative of (11) with respect to $\mathbf{Z}$ and setting it to zero, we have the closed form solution of $\mathbf{Z}_{t+1}$ as follows:

$$\mathbf{Z}_{t+1} = \left(\mathbf{X}^\top \mathbf{X} + \gamma \mathbf{G}_t + \alpha \mathbf{X}^\top \mathbf{L}\mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{X}. \tag{12}$$

Since the l_2-norm of $\mathbf{z}^i$ could be very close to zero, we often introduce a sufficiently small value ε to avoid the overflow error during the iteration process and define $g_t^i = 1/\max(2\|\mathbf{z}_t^i\|_2, \varepsilon)$.

2) Fix $\mathbf{Z}$ and $\boldsymbol{\omega}$, Update $\mathbf{S}$: When $\mathbf{Z}$ and $\boldsymbol{\omega}$ are fixed, $\mathbf{S}$ can be updated via solving following problem:

$$\begin{aligned} &\min_{\mathbf{S}} \sum_{v=1}^{\tau} \boldsymbol{\omega}_v ||\mathbf{S} - \mathbf{S}^v||_F^2 + \alpha \mathrm{Tr}\left(\mathbf{Z}^\top \mathbf{X}^\top \mathbf{L}\mathbf{X}\mathbf{Z}\right) \\ &\text{s.t. } S_{ij} \geq 0, \mathbf{s}_i^\top \mathbf{1}_N = 1. \end{aligned} \tag{13}$$

Since problem (13) is independent for different $\mathbf{s}_i$, we can solve each $\mathbf{s}_i$ separately via optimizing the following problem:

$$\begin{aligned} &\min_{\mathbf{s}_j} \sum_{v=1}^{\tau} \sum_{i=1}^{N} \boldsymbol{\omega}_v \left(\mathbf{s}_{ij} - \mathbf{s}_{ij}^v\right)^2 + \alpha \sum_{i,j=1}^{N} ||\mathbf{p}^i - \mathbf{p}^j||_2^2 \mathbf{s}_{ij} \\ &\text{s.t. } S_{ij} \geq 0, \mathbf{s}_i{}^\top \mathbf{1}_N = 1 \end{aligned} \tag{14}$$

where $\mathbf{p}^i$ and $\mathbf{p}^j$ represent the ith and jth row of $\mathbf{X}\mathbf{Z}$, respectively. By denoting $d_{ij} = ||\mathbf{p}^i - \mathbf{p}^j||_2^2$, then the following equation can be rewritten as:

$$\begin{aligned} &\min_{\mathbf{s}_j} \sum_{v=1}^{\tau} \sum_{i=1}^{N} \boldsymbol{\omega}_v \left(\mathbf{s}_{ij} - \mathbf{s}_{ij}^v\right)^2 + \alpha \sum_{i,j=1}^{N} d_{ij} \mathbf{s}_{ij} \\ &\text{s.t. } S_{ij} \geq 0, \mathbf{s}_i{}^\top \mathbf{1}_N = 1 \end{aligned} \tag{15}$$

which can be efficiently solved via the Lagrangian function and Karush–Kuhn–Tucker (KKT) conditions [71].

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3) Fix $\mathbf{Z}$ *and* $\mathbf{S}$*, Update* $\boldsymbol{\omega}$*:* When $\mathbf{Z}$ and $\mathbf{S}$ are fixed, $\boldsymbol{\omega}$ can be updated via solving following problem:

$$\min_{\boldsymbol{\omega}_v} \sum_{v=1}^{\tau} \boldsymbol{\omega}_v ||\mathbf{S} - \mathbf{S}^v||_F^2 \quad \text{s.t.} \quad \boldsymbol{\omega}^\top \mathbf{1}_\tau = 1. \tag{16}$$

Based on Theorem 1, $\boldsymbol{\omega}_v$ can be determined automatically.

Theorem 1: When the weights $\boldsymbol{\omega}_1, \boldsymbol{\omega}_2, \cdots, \boldsymbol{\omega}_\tau$ are given, problem (7) is equal to the following problem:

$$\min_{\mathbf{S}} \sum_{v=1}^{\tau} \sqrt{||\mathbf{S} - \mathbf{S}^v||_F^2}, \quad \text{s.t.} \quad S_{ij} \geq 0, \mathbf{s}_i{}^\top \mathbf{1}_N = 1. \tag{17}$$

Proof: In order to solve (17), we construct its corresponding Lagrange function as follows:

$$\mathcal{L}(\mathbf{S}, \Phi) = \sum_{v=1}^{\tau} \sqrt{||\mathbf{S} - \mathbf{S}^v||_F^2} + f(\Phi, \mathbf{S}) \tag{18}$$

where Φ is the Lagrange multiplier and $f(\Phi, \mathbf{S})$ is a unified regularization term derived from the constraints.

Taking the derivative of $\mathcal{L}(\mathbf{S}, \Phi)$ with respect to $\mathbf{S}$ and setting the derivative to zero, we have

$$\sum_{v=1}^{\tau} \boldsymbol{\omega}_v \frac{\partial ||\mathbf{S} - \mathbf{S}^v||_F^2}{\partial \mathbf{S}} + \frac{\partial f(\Phi, \mathbf{S})}{\partial \mathbf{S}} = 0 \tag{19}$$

where

$$\boldsymbol{\omega}_v = \frac{1}{2\sqrt{||\mathbf{S} - \mathbf{S}^v||_F^2}}. \tag{20}$$

Therefore, if $\boldsymbol{\omega}_v$ is given, the derivative of the Lagrange function for (16) is equal to (19), which indicates that the optimization problem described by (16) is equal to (17), and $\boldsymbol{\omega}_v$ can be calculated by (20). □

In a nutshell, the whole process for solving the proposed model can be summarized by Algorithm 1.

Algorithm 1 S^{4}P

Input: HSI cube P, the number of selected bands K, the number of superpixels N, hyper-parameters α and γ.
Output: The obtained band subset.
1: Segment the HSI cube P into N regions with using PCA and ERS.
2: Construct the superpixel level feature matrix $\mathbf{X}$.
3: Initialize $\boldsymbol{\omega}$, $\mathbf{S}$.
4: **while not converged do**
5: Update $\mathbf{Z}$ via (12).
6: Update $\mathbf{S}$ via (13).
7: Update $\boldsymbol{\omega}$ via (16).
8: **end while**
9: Conduct the spectral clustering on $\mathbf{Z}$ to generate multiple clusters, and the band closest to the cluster center is selected.
10: **Return the band subset.**

D. Algorithm Computational Complexity Analysis

The time complexity of our proposed S^{4}P consists of two parts: image preprocessing (including PCA and superpixel segmentation) and optimization algorithm for solving problem 9. As to PCA, the time complexity of computing the covariance matrix and eigen-decomposition are $\mathcal{O}(B^2WH)$ and $\mathcal{O}(B^3)$, respectively. As to the ERS superpixel segmentation algorithm, its time complexity is $\mathcal{O}(WH\log(WH))$. As presented in Section III-C, there are three steps to solve the resulting optimization problem. For updating $\mathbf{Z}$, the major computational cost is the inverse of a $B \times B$ matrix, which is $\mathcal{O}(B^3)$. For updating $\mathbf{S}$, each row $\mathbf{s}_i$ can be efficiently solved via the Lagrangian function and KKT conditions, and the whole computational cost is $\mathcal{O}(N)$. For updating $\boldsymbol{\omega}$, the computational cost is the Frobenius norm of a $N \times N$ matrix, which is negligible. Therefore, the whole computational cost of the proposed S^{4}P is $\mathcal{O}(B^2WH) + \mathcal{O}(WH\log(WH)) + t(\mathcal{O}(B^3) + \mathcal{O}(N))$ with t denoting the iteration times of Algorithm 1. It should be noted that B and N represent the band number of original hyperspectral data and the number of segmented superpixels, respectively, and their values are both around 100–300 in practice. As a result, compared to other previous methods which stack each band into a high-dimensional feature vector, our proposed algorithm is very efficient.

In order to validate the effectiveness of our proposed S^{4}P, we report the computational time of all competitors on three datasets. For a fair comparison, all nondeep learning-based methods are tested on a PC workstation with Intel Core i5-3470 CPU processor and 16 GB RAM while deep learning-based methods are tested with an extra single NVIDIA Titan V GPU card. Table I shows the execution time of different methods with ten selected bands on different datasets. As can be seen from the results, although our proposed S^{4}P does not perform the best, it still outperforms some nondeep learning-based methods such as similarity-based ranking (SR-SSIM) and MSGCF.

IV. Experiments

In this section, we present experimental results conducted on three public HSI datasets to validate the efficacy of the proposed method. In addition, parameter sensitivity and algorithm convergence property analysis are also discussed.

A. Datasets

The three datasets used in our experiments are the Indian Pines scene, the Pavia University scene, and the Botswana scene. The details of different datasets are summarized as follows.

Indian Pines Scene, which was captured by the AVIRIS sensor in 1992. There are a total of 224 spectral bands from 0.4 to 2.5 μm, and the spatial size of each band is 145×145 pixels. There are 16 different classes of land cover objects in this dataset. By discarding 20 spectral bands (indexed 104–108, 150–163, and 220) that are heavily interfered with by water absorption, the rest 200 bands are used for experiments.

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TABLE I
RUNNING TIME COMPARISON OF DIFFERENT BAND SELECTION METHODS ON DIFFERENT DATASETS WITH TEN SELECTED BANDS

	UBS	EFDPC	OPBS	OCF	ASPS-IE	FNGBS	SR-SSIM	MSGCF	SRLSOA	RLSBS	S4P
Indian Pines	0.01	0.15	0.46	0.99	0.53	0.39	1.34	2.63	3.72	4.24	2.22
Pavia University	0.01	0.32	0.93	1.32	0.61	1.57	2.27	2.94	5.68	6.71	2.84
Botswana	0.03	0.97	2.75	2.64	1.96	3.95	3.26	4.25	7.23	8.03	4.66

Pavia University Scene, which was captured over Pavia by the ROSIS sensor in 2002. There are a total of 115 bands in the original data and the spatial resolution is 1.3 m. There are nine classes of land cover objects in this scene, and the spatial size of each band is 610×340. By removing 12 noisy bands, the rest 103 bands are used.

Botswana Scene, which was obtained by NASA EO-1 satellite sensor over Okavango Delta, Botswana between 2001 and 2004. There are a total of 242 spectral bands with the size of 1476×256 pixels in this dataset. In order to keep consistent with previous works, we also select 145 bands for experiments by discarding other 97 spectral bands with heavy noise.

B. Compared Methods

In order to validate the superiority of our proposed S^4P, we compare it with other ten state-of-the-art unsupervised hyperspectral band selection methods, which include as follows.

1) UBS [28]: Constrained Band Selection for hyperspectral imagery, in which the hyperspectral cube is divided into multiple subcubes at equal width based on the required number of selected bands, then each segmentation point is selected as the discriminative band.

2) EFDPC [72]: A Novel ranking-based clustering approach for hyperspectral band Selection, which enhances the fast density-peak-based clustering algorithm to make it suitable for hyperspectral band selection.

3) OCF [33]: Optimal clustering framework for hyperspectral band selection. This method proposes an optimal clustering framework that can obtain the optimal clustering result for a particular form of objective function under a reasonable rank constraint.

4) OPBS [73]: A geometry-based band selection approach for HSI analysis, which combines the maximum ellipsoid volume with the sequential forward search and proposes a new unsupervised hyperspectral band selection.

5) ASPS_IE [16]: Hyperspectral band selection via adaptive subspace partition strategy, which selects discriminative bands via an adaptive subspace partition strategy and information entropy.

6) FNGBS [36]: A fast neighborhood grouping method for hyperspectral band selection, which divides the original hyperspectral cube into several subsets based on neighborhood grouping, then the band with the highest local density and information entropy in each band subset is selected.

7) SR-SSIM [26]: A similarity-based ranking method for hyperspectral band selection, which uses the SSIM indicator to measure the similarity between different bands.

8) MSGCF [69]: An MSGCF for hyperspectral band selection, which mainly employs the superpixel-level distance to measure the correlations among spectral bands.

9) SRLSOA [74]: Self-representation learning with sparse 1-D-operational autoencoder for HSI band selection, which aims to learn a representation domain from the sparse data space.

10) RLSBS [75]: Deep reinforcement learning for semisupervised hyperspectral band selection. This method regards the band selection as a reinforcement learning problem and each search space can be treated as one of the feasible band subsets.

C. Experimental Setup

In our experiments, we use HSI classification performance obtained from the selected band subsets to measure the efficacy of different band selection methods, and three metrics including the overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa) are used as the evaluation criteria. Three typical traditional classifiers, including k-nearest neighborhood (KNN), LDA, and support vector machine (SVM) are used for classification evaluation. Since the optimal number of selected bands is unknown and yet hard to determine for different datasets, we test the classification accuracy of different classifiers on different datasets by varying the number of selected bands from 5 to 50 with step 5. Specifically, we first solve problem (9) by using Algorithm 1 to get the band coefficient matrix $\mathbf{Z}$, then spectral clustering is conducted on $\mathbf{Z}$ to generate multiple clusters corresponding to a certain number of selected bands, and the band closest to the cluster center is selected. The parameter k of KNN is set to 3 for all experiments. As to SVM classifier, the RBF kernel is used, and we set the penalty C and gamma to 1e4 and 0.5, respectively. In the classification phase, for each number of selected bands, since we randomly sample 10% samples from each class for training for each dataset, and the rest 90% samples are used for testing. In order to reduce the sampling random bias, each classification algorithm is run ten times and the average classification accuracies are reported. There are two parameters in our proposed model, i.e., α and γ, we tune their values in the range $\{10^{-4}, 10^{-2}, \ldots, 10^{4}\}$. As to other compared methods, we tune the parameters according to the suggestions in the original papers. Since different HSIs could be of different sizes, we need to adaptively set the number of segmented superpixels for different datasets. Therefore, similar to previous hyperspectral classification work [76], we determine the number of segmented superpixels as follows:

$$N = T \times \frac{\mathcal{N}_z}{\mathcal{N}} \tag{21}$$

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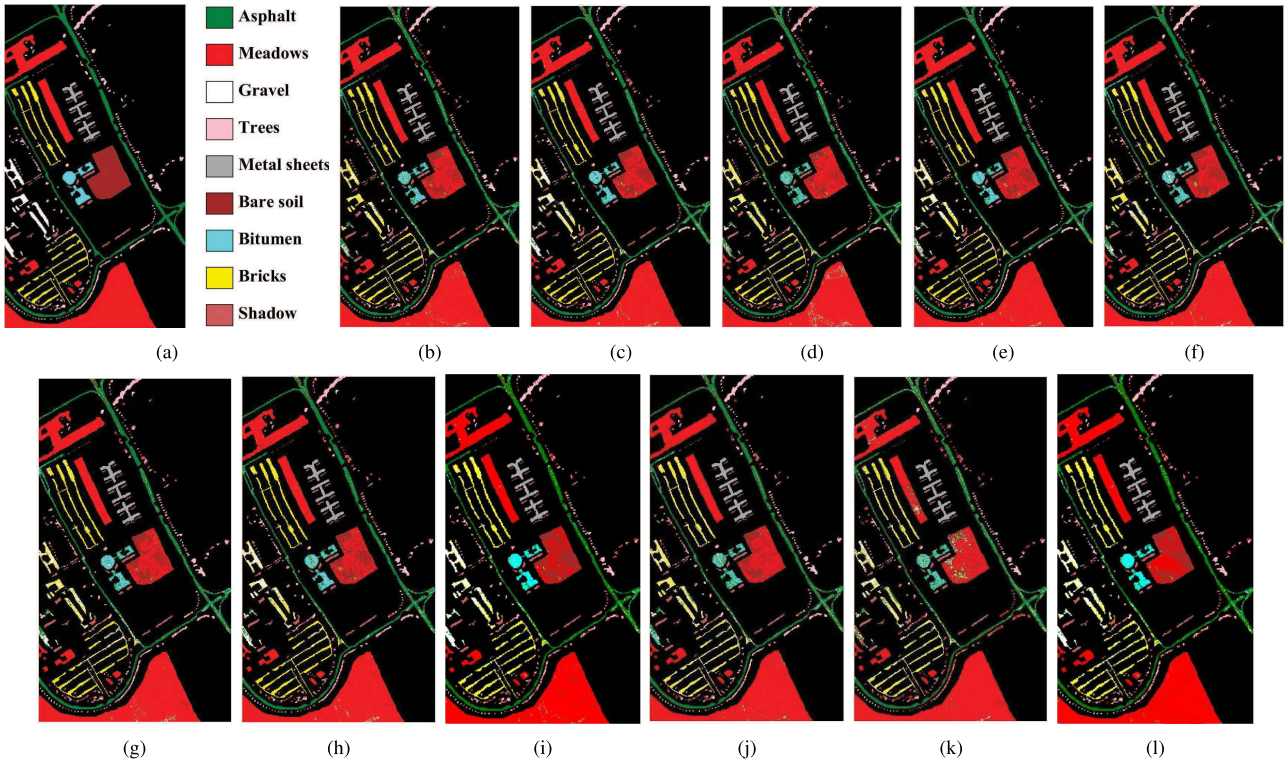


Fig. 2. Visual classification maps of all compared methods on the Pavia University dataset using KNN classifier. (a) Ground truth. (b) UBS. (c) EFDPC. (d) OPBS. (e) OCF. (f) ASPS-IE. (g) FNGBS. (h) SR-SSIM. (i) MSGCF. (j) SRLSOA. (k) RLSBS. (l) S^{4}P.

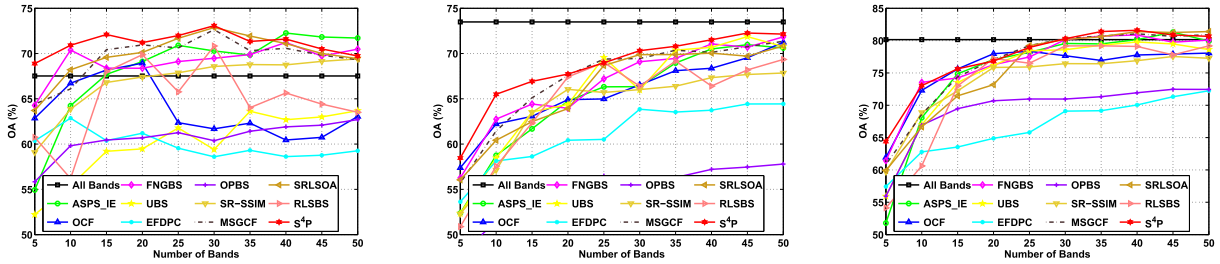


Fig. 3. OA of three different classifiers with varied numbers of selected bands on the Indian Pines dataset. (a) OA obtained by KNN. (b) OA obtained by SVM. (c) OA obtained by LDA.

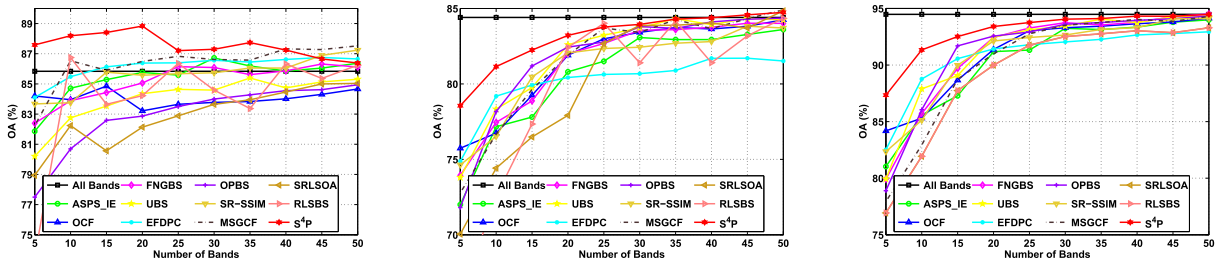


Fig. 4. OA of three different classifiers with varied numbers of selected bands on the Pavia University dataset. (a) OA obtained by KNN. (b) OA obtained by SVM. (c) OA obtained by LDA.

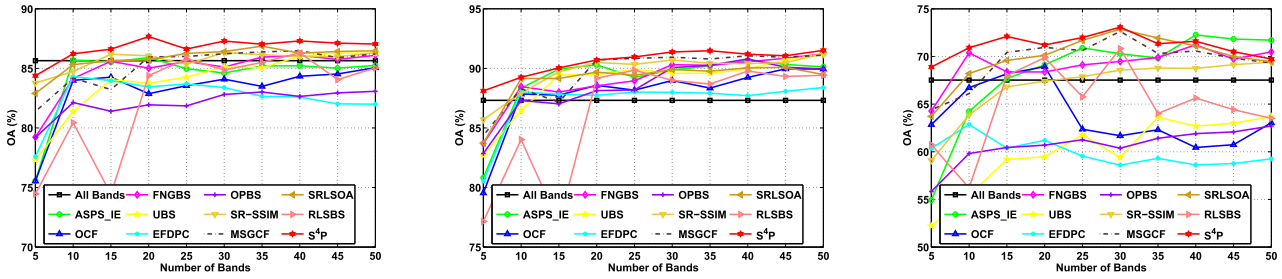
where $\mathcal{N}_z$ denotes the number of nonzero values at the edge of the first principal component of HSIs, $\mathcal{N}$ denotes the total number of pixels of each band, and T is a fixed number which we set to 2000.

D. Experimental Results

Since it is difficult to determine the optimal number of bands for a certain hyperspectral dataset, in Figs. 3–5, we plot the OA curves for each dataset with an interval of five bands. As can be seen, our proposed S^{4}P obtains higher OA than other competitors in nearly all of the numbers of selected bands on different datasets.

In addition, the proposed method can obtain better classification results than by using all of the original bands in most cases, which can be attributed to the high redundancy present in the latter. It is well known that original HSIs often

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(a) (b) (c)

Fig. 5. OA of three different classifiers with varied numbers of selected bands on the Botswana dataset. (a) OA obtained by KNN. (b) OA obtained by SVM. (c) OA obtained by LDA.

TABLE II

CLASSIFICATION RESULTS OF DIFFERENT METHODS WITH THREE METRICS ON DIFFERENT DATASETS

Datasets	Classifier	Metrics	UBS	EFDPC	OPBS	OCF	ASPS-IE	FNGBS	SR-SSIM	MSGCF	SRLSOA	RLSBS	S^4P
Indian Pines	KNN	OA	59.68±0.56	58.59±0.64	60.37±0.95	61.89±0.45	69.96±0.65	69.46±0.49	68.57±0.62	71.42±0.63	**73.12±0.69**	70.83±0.54	73.07±0.70
		AA	47.36±0.71	44.31±1.35	48.41±0.66	49.16±1.31	55.99±1.44	56.15±1.94	57.15±1.78	58.21±1.82	**60.07±1.60**	58.00±1.82	58.46±2.72
		Kappa	57.11±0.58	56.00±0.63	57.85±0.95	59.31±0.44	67.67±0.67	67.08±0.51	66.24±0.65	68.91±0.62	**70.91±0.70**	68.54±0.54	70.81±0.75
	SVM	OA	78.58±0.62	69.09±0.87	70.96±0.91	77.32±0.43	79.28±0.41	**80.51±0.49**	76.44±0.51	79.25±0.65	80.18±0.53	79.18±0.75	80.29±0.42
		AA	72.62±1.85	55.82±1.24	64.48±1.53	72.45±2.58	75.68±1.75	77.26±2.17	72.99±1.62	76.84±0.56	77.00±1.65	75.35±2.37	**77.55±1.17**
		Kappa	76.68±0.65	66.52±0.94	68.74±0.92	75.35±0.46	77.42±0.44	78.71±0.52	74.45±0.54	77.52±0.50	78.38±0.57	77.30±0.79	**78.49±0.44**
	LDA	OA	66.62±0.78	63.85±0.79	56.36±0.68	66.45±0.52	66.14±0.90	70.20±0.85	66.01±0.70	68.74±0.59	69.88±0.82	66.40±0.78	**70.32±0.22**
		AA	61.94±2.29	57.39±2.53	50.89±2.01	60.86±1.31	60.61±1.53	66.86±2.88	63.37±0.67	66.56±0.37	**67.76±1.81**	64.22±1.81	66.94±0.40
		Kappa	64.26±0.81	61.54±0.84	53.55±0.68	64.07±0.51	63.87±0.91	68.03±0.89	63.54±0.69	66.57±0.62	67.74±0.83	64.08±0.79	**68.18±0.60**
Pavia University	KNN	OA	84.61±0.22	86.47±0.13	84.00±0.31	83.86±0.20	86.64±0.22	86.57±0.17	85.74±0.20	84.37±0.21	83.64±0.16	84.59±0.16	**87.30±0.17**
		AA	80.96±0.23	83.51±0.25	79.98±0.47	79.58±0.22	83.69±0.37	83.48±0.37	82.51±0.24	80.37±0.23	79.45±0.25	80.49±0.26	**84.27±0.15**
		Kappa	80.50±0.26	82.77±0.16	79.73±0.37	79.55±0.24	82.98±0.28	82.89±0.22	81.85±0.24	78.96±0.18	79.32±0.19	80.44±0.20	**83.77±0.19**
	SVM	OA	93.39±0.18	92.04±0.14	93.60±0.13	93.32±0.10	93.19±0.01	93.72±0.01	92.65±0.13	91.78±0.13	92.99±0.12	92.89±0.12	**94.06±0.12**
		AA	91.11±0.33	89.26±0.35	91.33±0.27	91.03±0.21	90.91±0.18	91.49±0.27	90.29±0.21	90.67±0.31	90.38±0.26	90.65±0.22	**91.80±0.47**
		Kappa	91.44±0.24	89.72±0.18	91.72±0.17	91.35±0.13	91.19±0.12	91.86±0.11	90.51±0.16	90.75±0.16	90.93±0.16	90.82±0.16	**92.28±0.15**
	LDA	OA	83.58±0.25	80.65±0.24	83.80±0.19	83.46±0.15	82.99±0.30	83.52±0.32	82.43±0.31	82.96±0.22	83.21±0.29	82.82±0.23	**83.94±0.21**
		AA	77.95±0.39	73.41±0.22	77.97±0.25	77.51±0.32	77.13±0.36	77.99±0.35	76.58±0.40	76.87±0.42	76.99±0.52	77.38±0.40	**78.52±0.33**
		Kappa	79.45±0.31	75.95±0.27	79.71±0.21	79.31±0.17	78.73±0.34	79.39±0.37	78.12±0.36	78.82±0.27	78.98±0.34	78.63±0.28	**79.84±0.25**
Botswana	KNN	OA	85.25±0.88	83.39±0.82	82.81±0.43	84.07±0.80	84.29±0.81	84.82±1.23	86.33±0.81	85.39±0.72	86.41±0.63	84.87±0.95	**87.29±0.66**
		AA	86.37±1.08	84.36±1.02	83.90±0.82	85.22±0.70	85.56±0.82	85.89±2.01	87.43±0.68	87.08±0.97	87.43±0.67	86.03±0.90	**88.31±1.04**
		Kappa	84.36±0.92	82.43±0.85	81.82±0.45	83.13±0.83	83.36±0.84	83.91±1.29	85.49±0.85	84.17±0.73	85.57±0.66	83.96±0.99	**86.49±0.70**
	SVM	OA	89.58±0.51	88.20±0.78	89.64±0.79	88.79±0.66	89.46±0.84	89.54±0.70	90.27±0.64	88.97±0.57	90.30±0.69	88.70±0.71	**91.68±0.23**
		AA	90.71±0.50	89.31±0.63	90.56±0.75	89.88±0.72	90.43±0.66	90.66±0.66	91.21±0.68	90.36±0.59	91.20±0.66	89.94±0.64	**92.63±0.41**
		Kappa	88.90±0.54	87.44±0.81	88.96±0.83	88.06±0.69	88.77±0.88	88.85±0.73	89.61±0.67	88.83±0.60	89.65±0.72	87.97±0.75	**91.11±0.25**
	LDA	OA	90.06±0.71	87.99±0.59	90.06±0.97	88.53±0.80	90.00±0.65	90.22±0.36	90.70±0.65	89.06±0.57	89.90±0.79	88.99±0.61	**91.38±0.34**
		AA	90.76±1.04	89.10±0.49	90.98±1.09	89.55±0.96	90.72±0.93	90.84±0.59	91.59±0.73	90.34±0.68	90.67±1.00	89.83±0.63	**92.13±0.45**
		Kappa	89.39±0.75	87.22±0.62	89.40±1.02	87.79±0.84	89.34±0.68	89.56±0.38	90.07±0.68	88.59±0.58	89.23±0.84	88.27±0.64	**90.78±0.36**

TABLE III

OA WITH SELECTING DIFFERENT NUMBERS OF SUPERPIXELS BY USING KNN CLASSIFIER (%)

Datasets	T	N	5	10	15	20	25	30	35	40	45	50
Indian Pines	500	41	67.58	68.71	67.36	68.64	68.45	67.20	67.70	67.20	67.62	67.99
	1000	83	68.09	68.20	70.04	69.89	70.54	70.51	69.75	70.68	70.76	71.37
	2000	165	68.89	70.92	72.10	71.21	71.99	73.07	71.34	71.60	70.51	69.73
	3000	248	67.66	68.93	70.38	71.78	72.43	72.46	72.16	73.30	72.65	72.08
Pavia University	500	39	88.00	88.92	86.66	86.72	86.81	86.75	86.71	86.75	86.22	86.40
	1000	78	88.14	88.81	88.79	87.94	87.67	86.13	86.45	86.81	86.58	86.68
	2000	155	87.59	88.19	88.41	88.83	87.21	87.30	87.74	87.24	86.66	86.38
	3000	233	87.76	88.45	88.36	88.58	87.36	87.50	87.45	87.24	86.63	86.79
Botswana	500	45	85.42	85.93	86.05	87.17	86.63	86.91	87.34	87.33	87.71	87.27
	1000	90	84.21	86.06	86.66	86.52	86.76	86.78	86.52	86.67	87.51	87.18
	2000	180	84.38	86.22	86.61	87.67	86.62	87.29	87.03	87.30	87.12	87.04
	3000	270	83.95	86.71	86.47	87.03	86.62	87.11	86.96	87.33	86.72	86.77

contain noise and a significant amount of redundant information, both of which can adversely affect final classification accuracy. Increasing the number of selected bands can indeed help enhance the classification performance to some extent; however, this improvement is constrained by the presence of noise and redundant information. Consequently, full bands cannot consistently maintain optimal performance in HSIs classification, necessitating the need for band selection.

In order to give a more compact comparison, we show the averaged OA of each method on each dataset with varied

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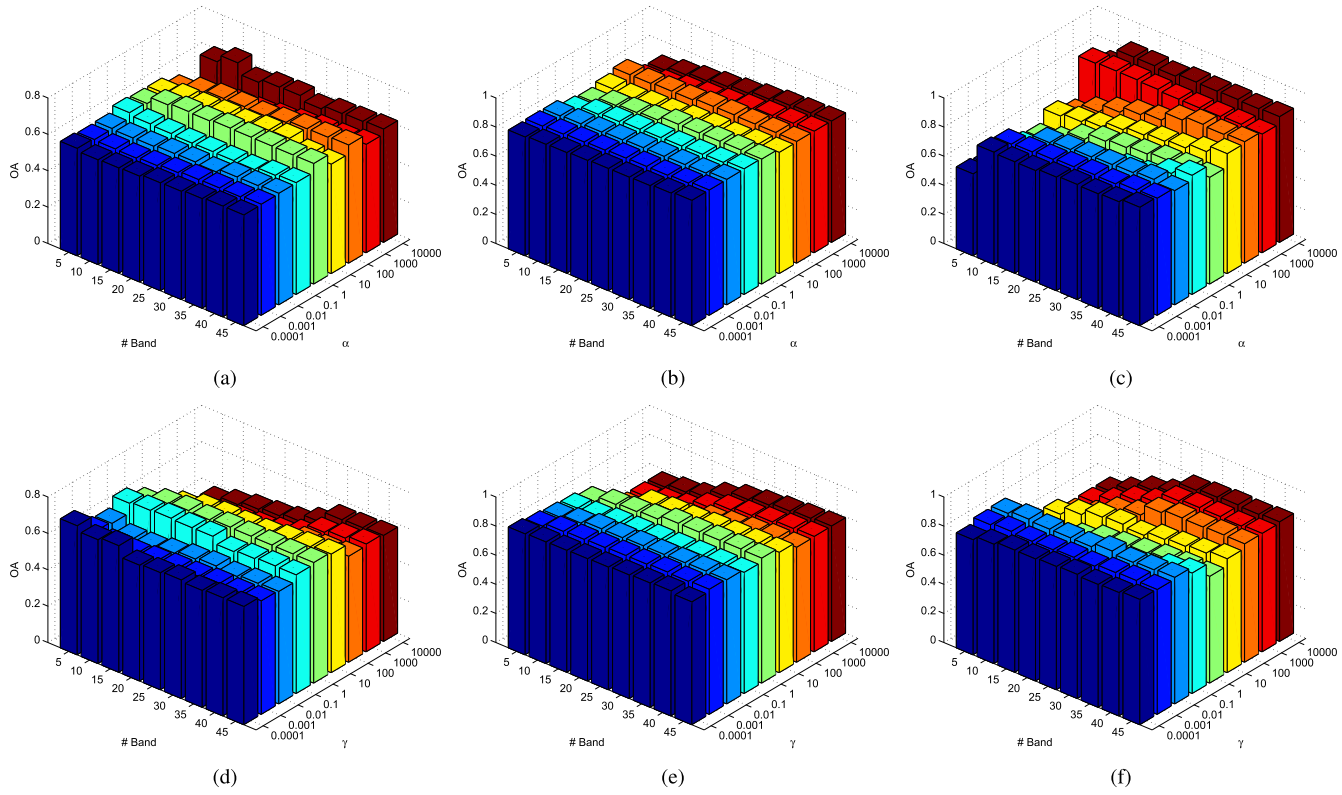


Fig. 6. Parameter sensitivity analysis of OA with KNN classifier on three different datasets. (a) Indian Pines ($\gamma = 1$). (b) Pavia University ($\gamma = 1$). (c) Botswana ($\gamma = 1$). (d) Indian Pines ($\alpha = 1$). (e) Pavia University ($\alpha = 1$). (f) Botswana ($\alpha = 1$).

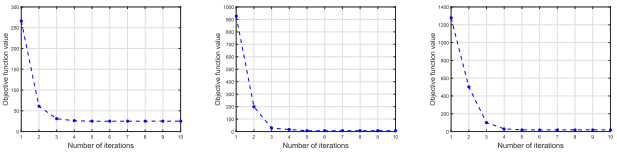


Fig. 7. Convergence curves of the proposed algorithm on three different datasets ($\alpha = 0.0001, \gamma = 0.0001$). (a) Indian Pines. (b) Pavia University. (c) Botswana.

numbers of selected bands in Table II. As shown by the results, our proposed S^{4}P consistently outperforms other methods in terms of different classifiers on Pavia University and Botswana datasets. As to the Indian Pines dataset, although our proposed S^{4}P does not perform the best, it can still get competitive performance than the best one.

Additionally, taking the Pavia University dataset as an example, the visual classification results of different methods obtained from the top five selected bands are presented in Fig. 2. From Fig. 2, we can find that the classification map of the proposed S^{4}P is cleaner than that of other competitors. For example, red areas classified as the meadows are less influenced by the other categories in the below Fig. 2(l), which verifies the effectiveness of the proposed method for band selection, and the selected bands are better for following hyperspectral classification.

E. Parameter Sensitivity Analysis

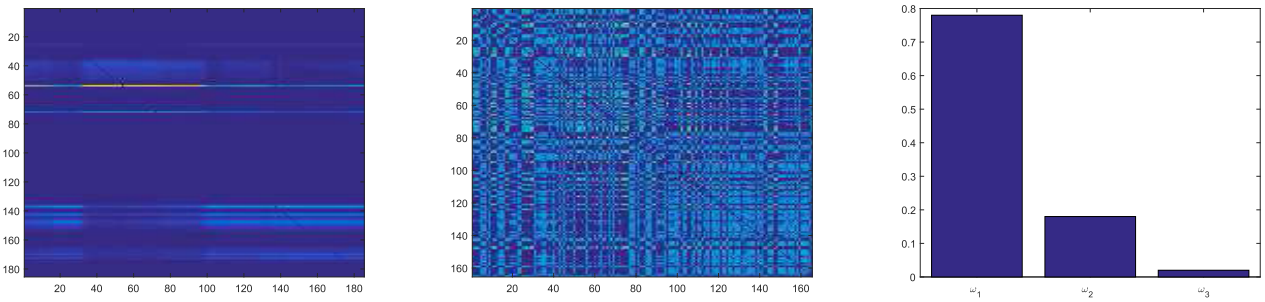
As shown by (9), two parameters including α and γ need to be tuned in our experiments. In order to study the sensitivity of S^{4}P with regard to the parameters, we conduct experiments by fixing one parameter and varying the rest. Firstly, we fix $\gamma = 1$ and vary α. Then, we fix $\alpha = 1$ and vary γ. In Figs. 6, we plot the OA values with different numbers of selected bands and parameter values of different datasets. As can be seen from the results, when the number of selected bands reaches 30, the final classification performance stays stable, which is also consistent with the results shown in Figs. 3–5. When $\alpha = 100$ and $\gamma = 0.01$, the proposed S^{4}P model can get the best results. In addition, it is not very sensitive to different α. Although the performance is sensitive to different γ to some extent, it can still obtain stable results when γ varies in a certain range.

Furthermore, as aforementioned, it is difficult to determine the desired number of superpixels in practice for different datasets, and we simply set T to 2000 in this article. To further study the impact of different numbers of superpixels on the final hyperspectral classification performance, we set T to 500, 1000, 2000, and 3000, respectively, and the corresponding classification results of the proposed method are shown in Table III. As seen in Table III, we find that the proposed method is not sensitive to T, and it can achieve relatively superior classification performance on all hyperspectral datasets when $T = 2000$.

F. Visualization of Learned $\mathbf{Z}$, $\mathbf{S}$, and $\boldsymbol{\omega}$

As formulated by (9), $\mathbf{Z} \in \mathbb{R}^{B\times B}$ denotes the coefficient matrix with its ith row expressing the representation capability of original ith band for reconstructing other bands. Therefore, $\mathbf{Z}$ should be row-sparse since a certain band should be

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(a) (b) (c)

Fig. 8. Visualization of learned $\mathbf{Z}$, $\mathbf{S}$, and $\boldsymbol{\omega}$ on the Indian Pines Scene dataset ($\alpha = \gamma = 1$). (a) Visualization of $\mathbf{Z}$. (b) Visualization of $\mathbf{S}$. (c) Visualization of $\boldsymbol{\omega}$.

reconstructed by only several bands, which is consistent with the concept of dictionary learning [77]. In order to demonstrate the row-sparsity property of the learned $\mathbf{Z}$, we visualize $\mathbf{Z}$ learned on the Indian Pines Scene dataset in Fig. 8(a). As can be seen, the row-sparsity structure can be well learned by the proposed model. The similarity matrix of different bands and the weights of different principle components are also shown in Fig. 8(b) and (c), respectively. We can observe that the relationship of different bands can be well reflected by the fused $\mathbf{S}$. In addition, the weight of the first principle component is the largest, which is also consistent with traditional principle component analysis. Therefore, we only need the first several principal components for good performance, as well as reduced computational cost.

G. Convergence Property Validation

It is not easy to theoretically prove the convergence property of the proposed algorithm, but the convergence property of each step for solving the variables can be well guaranteed. We empirically plot the convergence curves of three datasets in Fig. 7 and the results show the strong convergence property of the proposed algorithm. The object function values on all the datasets decrease fast within five iteration times and stay stable along with more iterations.

V. Conclusion

In this article, we introduce a spatial and spectral structure preserved self-representation model for unsupervised hyperspectral band selection, which fully takes both the spatial and spectral information of the original hyperspectral data cube into consideration. An adaptive and weighted multiple graph fusion strategy is designed to learn a unified similarity graph between different regions of the 2-D hyperspectral slice for spatial structure preservation in the spectral self-representation space. Extensive experimental results with detailed analysis of different datasets demonstrate the superiority of the proposed method when compared with other competitors.

Following the previous works, the optimal number of segmented superpixels of each dataset is determined experimentally in our proposed method. Although this manner can achieve a satisfying performance on hyperspectral band selection, how to directly obtain the optimal number of superpixels is still a difficult problem for a new dataset without experiments, which is one of the main limitations of this article. Furthermore, similar to other band selection methods, how to determine the optimal number of selected bands is also a main challenge in the proposed method. Thus, we will take these challenges into consideration in the future.

References

[1] A. F. H. Goetz, "Three decades of hyperspectral remote sensing of the Earth: A personal view," *Remote Sens. Environ.*, vol. 113, pp. S5–S16, Sep. 2009.

[2] S. Huang, H. Zhang, and A. Pižurica, "Hybrid-hypergraph regularized multiview subspace clustering for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505816.

[3] D. Haboudane, "Hyperspectral vegetation indices and novel algorithms for predicting green LAI of crop canopies: Modeling and validation in the context of precision agriculture," *Remote Sens. Environ.*, vol. 90, no. 3, pp. 337–352, Apr. 2004.

[4] B. Gao et al., "Additional sampling layout optimization method for environmental quality grade classifications of farmland soil," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 12, pp. 5350–5358, Dec. 2017.

[5] M. H. Zadeh, M. H. Tangestani, F. V. Roldan, and I. Yusta, "Mineral exploration and alteration zone mapping using mixture tuned matched filtering approach on ASTER data at the central part of dehaj-sarduiyeh copper belt, SE Iran," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 1, pp. 284–289, Jan. 2014.

[6] C. Ye et al., "Landslide detection of hyperspectral remote sensing data based on deep learning with constrains," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 5047–5060, Dec. 2019.

[7] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 7, no. 2, pp. 118–139, Jun. 2019.

[8] C.-I. Chang, Q. Du, T.-L. Sun, and M. L. G. Althouse, "A joint band prioritization and band-decorrelation approach to band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 6, pp. 2631–2641, Jul. 1999.

[9] M. P. Uddin, M. A. Mamun, M. I. Afjal, and M. A. Hossain, "Information-theoretic feature selection with segmentation-based folded principal component analysis (PCA) for hyperspectral image classification," *Int. J. Remote Sens.*, vol. 42, no. 1, pp. 286–321, Jan. 2021.

[10] H. Du, H. Qi, X. Wang, R. Ramanath, and W. E. Snyder, "Band selection using independent component analysis for hyperspectral image processing," in *Proc. 32nd Appl. Imag. Pattern Recognit. Workshop*, 2003, pp. 93–98.

[11] J. Wang and C.-I. Chang, "Independent component analysis-based dimensionality reduction with applications in hyperspectral image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1586–1600, Jun. 2006.

[12] T. V. Bandos, L. Bruzzone, and G. Camps-Valls, "Classification of hyperspectral images with regularized linear discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 3, pp. 862–873, Mar. 2009.

[13] A. A. Green, M. Berman, P. Switzer, and M. D. Craig, "A transformation for ordering multispectral data in terms of image quality with implications for noise removal," *IEEE Trans. Geosci. Remote Sens.*, vol. GRS-26, no. 1, pp. 65–74, Jan. 1988.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:30:34 UTC from IEEE Xplore. Restrictions apply.

[14] H. Fu et al., "A novel band selection and spatial noise reduction method for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535713.

[15] J. Tschannerl et al., "MIMR-DGSA: Unsupervised hyperspectral band selection based on information theory and a modified discrete gravitational search algorithm," *Inf. Fusion*, vol. 51, pp. 189–200, Nov. 2019.

[16] Q. Wang, Q. Li, and X. Li, "Hyperspectral band selection via adaptive subspace partition strategy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4940–4950, Dec. 2019.

[17] H. Yang, Q. Du, H. Su, and Y. Sheng, "An efficient method for supervised hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 138–142, Jan. 2011.

[18] J. Feng, L. Jiao, F. Liu, T. Sun, and X. Zhang, "Mutual-information-based semi-supervised hyperspectral band selection with high discrimination, high information, and low redundancy," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2956–2969, May 2015.

[19] X. Cao, C. Wei, Y. Ge, J. Feng, J. Zhao, and L. Jiao, "Semi-supervised hyperspectral band selection based on dynamic classifier selection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1289–1298, Apr. 2019.

[20] Q. Du and H. Yang, "Similarity-based unsupervised band selection for hyperspectral image analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 564–568, Oct. 2008.

[21] S. Jia, Z. Ji, Y. Qian, and L. Shen, "Unsupervised band selection for hyperspectral imagery classification without manual band removal," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 531–543, Apr. 2012.

[22] Y. Yuan, X. Zheng, and X. Lu, "Discovering diverse subset for unsupervised hyperspectral band selection," *IEEE Trans. Image Process.*, vol. 26, no. 1, pp. 51–64, Jan. 2017.

[23] N. Keshava, "Distance metrics and band selection in hyperspectral processing with applications to material identification and spectral libraries," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 7, pp. 1552–1565, Jul. 2004.

[24] A. MartÌnez-UsÓMartinez-Uso, F. Pla, J. M. Sotoca, and P. GarcÌa-Sevilla, "Clustering-based hyperspectral band selection using information measures," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 4158–4171, Dec. 2007.

[25] X. Geng, K. Sun, L. Ji, and Y. Zhao, "A fast volume-gradient-based band selection method for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 7111–7119, Nov. 2014.

[26] B. Xu, X. Li, W. Hou, Y. Wang, and Y. Wei, "A similarity-based ranking method for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9585–9599, Nov. 2021.

[27] B. Guo, S. R. Gunn, R. I. Damper, and J. D. B. Nelson, "Band selection for hyperspectral image classification using mutual information," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 4, pp. 522–526, Oct. 2006.

[28] C.-I. Chang and S. Wang, "Constrained band selection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1575–1585, Jun. 2006.

[29] W. Chen et al., "MIMN-DPP: Maximum-information and minimum-noise determinantal point processes for unsupervised hyperspectral band selection," *Pattern Recognit.*, vol. 102, Jun. 2020, Art. no. 107213.

[30] H. Su, Q. Du, G. Chen, and P. Du, "Optimized hyperspectral band selection using particle swarm optimization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2659–2670, Jun. 2014.

[31] Y. Yuan, J. Lin, and Q. Wang, "Dual-clustering-based hyperspectral band selection by contextual analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1431–1445, Mar. 2016.

[32] F. Zhang, Q. Wang, and X. Li, "Hyperspectral image band selection via global optimal clustering," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 1–4.

[33] Q. Wang, F. Zhang, and X. Li, "Optimal clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5910–5922, Oct. 2018.

[34] Q. Li, Q. Wang, and X. Li, "An efficient clustering method for hyperspectral optimal band selection via shared nearest neighbor," *Remote Sens.*, vol. 11, no. 3, p. 350, Feb. 2019.

[35] Y. Zhao, Y. Yuan, and Q. Wang, "Fast spectral clustering for unsupervised hyperspectral image classification," *Remote Sens.*, vol. 11, no. 4, p. 399, Feb. 2019.

[36] Q. Wang, Q. Li, and X. Li, "A fast neighborhood grouping method for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5028–5039, Jun. 2021.

[37] J. Peng et al., "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 10, no. 1, pp. 10–43, Mar. 2022.

[38] W. Sun et al., "A multiscale spectral features graph fusion method for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5513712.

[39] C. Sui et al., "Unsupervised hyperspectral band selection with multi-graph integrated embedding and robust self-contained regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504815.

[40] J. Wang, H. Wang, Z. Ma, L. Wang, Q. Wang, and X. Li, "Unsupervised hyperspectral band selection based on hypergraph spectral clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 5509905.

[41] Y. Zhang, X. Wang, X. Jiang, and Y. Zhou, "Robust dual graph self-representation for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5538513.

[42] S. Huang, H. Zhang, and A. Pižurica, "Subspace clustering for hyperspectral images via dictionary learning with adaptive regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524017.

[43] S. Huang, H. Zhang, J. Xue, and A. Pižurica, "Heterogeneous regularization-based tensor subspace clustering for hyperspectral band selection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9259–9273, Nov. 2022.

[44] S. Huang, H. Zhang, and A. Pižurica, "A structural subspace clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5509515.

[45] C. Tang, Z. Li, J. Wang, X. Liu, W. Zhang, and E. Zhu, "Unified one-step multi-view spectral clustering," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 6, pp. 6449–6460, Jun. 2023.

[46] C. Tang, X. Zheng, W. Zhang, X. Liu, X. Zhu, and E. Zhu, "Unsupervised feature selection via multiple graph fusion and feature weight learning," *Sci. China Inf. Sci.*, vol. 66, no. 5, pp. 1–17, May 2023.

[47] J. Wang, C. Tang, Z. Wan, W. Zhang, K. Sun, and A. Y. Zomaya, "Efficient and effective one-step multiview clustering," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Mar. 14, 2023, doi: 10.1109/TNNLS.2023.3253246.

[48] J. Wang et al., "Region-aware hierarchical latent feature representation learning-guided clustering for hyperspectral band selection," *IEEE Trans. Cybern.*, vol. 53, no. 8, pp. 5250–5263, Aug. 2022.

[49] J. Wang et al., "Hyperspectral band selection via region-aware latent features fusion based clustering," *Inf. Fusion*, vol. 79, pp. 162–173, Mar. 2022.

[50] H. Sun et al., "Adaptive distance-based band hierarchy (ADBH) for effective hyperspectral band selection," *IEEE Trans. Cybern.*, vol. 52, no. 1, pp. 215–227, Jan. 2022.

[51] H. Sun, L. Zhang, J. Ren, and H. Huang, "Novel hyperbolic clustering-based band hierarchy (HCBH) for effective unsupervised band selection of hyperspectral images," *Pattern Recognit.*, vol. 130, Oct. 2022, Art. no. 108788.

[52] K. He et al., "A dual global–local attention network for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527613.

[53] H. Sun, L. Zhang, L. Wang, and H. Huang, "Stochastic gate-based autoencoder for unsupervised hyperspectral band selection," *Pattern Recognit.*, vol. 132, Dec. 2022, Art. no. 108969.

[54] C. Yu, S. Zhou, M. Song, B. Gong, E. Zhao, and C.-I. Chang, "Unsupervised hyperspectral band selection via hybrid graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530515.

[55] X. Zhang, W. Xie, Y. Li, J. Lei, Q. Du, and G. Yang, "Rank-aware generative adversarial network for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521812.

[56] H. Sun, J. Ren, H. Zhao, P. Yuen, and J. Tschannerl, "Novel gumbel-softmax trick enabled concrete autoencoder with entropy constraints for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506413.

[57] M. M. Elkholy, M. S. Mostafa, H. M. Ebeid, and M. Tolba, "Unsupervised hyperspectral band selection with deep autoencoder unmixing," *Int. J. Image Data Fusion*, vol. 13, no. 3, pp. 244–261, Jul. 2022.

[58] S. Jia, S. Jiang, S. Zhang, M. Xu, and X. Jia, "Graph-in-graph convolutional network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 20, 2022, doi: 10.1109/TNNLS.2022.3182715.

[59] W. Sun, L. Zhang, B. Du, W. Li, and Y. Mark Lai, "Band selection using improved sparse subspace clustering for hyperspectral imagery classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2784–2797, Jun. 2015.

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[60] Y. Zhang, X. Wang, X. Jiang, and Y. Zhou, "Marginalized graph self-representation for unsupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516712.
[61] H. Zhai, H. Zhang, L. Zhang, and P. Li, "Laplacian-regularized low-rank subspace clustering for hyperspectral image band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1723–1740, Mar. 2019.
[62] H. Zhang, J. Zou, and L. Zhang, "EMS-GCN: An end-to-end mix-hop superpixel-based graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526116.
[63] C. Zhao et al., "Hyperspectral image classification with multi-attention transformer and adaptive superpixel segmentation-based active learning," *IEEE Trans. Image Process.*, vol. 32, pp. 3606–3621, 2023.
[64] J. Li, H. Sheng, M. Xu, S. Liu, and Z. Zeng, "BAMS-FE: Band-by-band adaptive multiscale superpixel feature extraction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5517015.
[65] T. Chen, Y. Liu, Y. Zhang, B. Du, and A. Plaza, "Superpixel-based collaborative and low-rank regularization for sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5529216.
[66] W. Cai, J. Jiang, and J. Qian, "Large-scale hyperspectral image restoration via a superpixel distributed algorithm based on graph signal processing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503517.
[67] M.-Y. Liu, O. Tuzel, S. Ramalingam, and R. Chellappa, "Entropy rate superpixel segmentation," in *Proc. CVPR*, Jun. 2011, pp. 2097–2104.
[68] S. Wold, K. Esbensen, and P. Geladi, "Principal component analysis," *Chemometrics Intell. Lab. Syst.*, vol. 2, nos. 1–3, pp. 37–52, 1987.
[69] S. Jia et al., "A multiscale superpixel-level group clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523418.
[70] H. Xu, W. He, L. Zhang, and H. Zhang, "Unsupervised spectral–spatial semantic feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526714.
[71] H. Wang, Y. Yang, and B. Liu, "GMC: Graph-based multi-view clustering," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 6, pp. 1116–1129, Jun. 2020.
[72] S. Jia, G. Tang, J. Zhu, and Q. Li, "A novel ranking-based clustering approach for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 88–102, Jan. 2016.
[73] W. Zhang, X. Li, Y. Dou, and L. Zhao, "A geometry-based band selection approach for hyperspectral image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4318–4333, Aug. 2018.
[74] M. Ahishali, S. Kiranyaz, I. Ahmad, and M. Gabbouj, "SRL-SOA: Self-representation learning with sparse 1D-operational autoencoder for hyperspectral image band selection," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2022, pp. 2296–2300.
[75] J. Feng et al., "Deep reinforcement learning for semisupervised hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501719.
[76] J. Jiang, J. Ma, Z. Wang, C. Chen, and X. Liu, "Hyperspectral image classification in the presence of noisy labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 851–865, Feb. 2019.
[77] I. Tošić and P. Frossard, "Dictionary learning," *IEEE Signal Process. Mag.*, vol. 28, no. 2, pp. 27–38, Mar. 2011.



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